Tourist

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

The Construction Kingdom is a famous tourist attraction, attracting thousands of tourists every year. The kingdom consists of n towns and m directed roads. Each road has a length of 1. It is guaranteed that among the initial m roads, there are no duplicate roads and no road from a town to itself.

Later, due to the continuous prosperity of tourism, the Construction Kingdom decided to expand the roads. Starting from day 1, over the next k days, a new directed road from x to y will be built each day. Due to improper planning, it may happen that an already existing road is built, or a road from a town to itself is built.

Fengmi is a free-spirited tourist. She has q travel wishes and a desired distance w. Each wish has a start s and an end t (s_i and t_i may be the same). She wants to know, for each wish, the earliest day on which she can start from town s, and under the condition that she is allowed to pass through towns and roads multiple times, exactly travel a distance of w to reach town t, or report that it is impossible. Please help her solve this problem.

Input

The first line contains two integers n and m ($2 \le n \le 200$, $1 \le m \le n(n-1)$) — the number of towns and the number of initial roads.

Each of the next m lines contains two integers u_i and v_i ($1 \le u_i, v_i \le n$, $u_i \ne v_i$), indicating a directed road from u_i to v_i.

The next line contains an integer k ($1 \le k \le n^2$) — the number of roads to be built in the following days.

Each of the next k lines contains two integers x_i and y_i ($1 \le x_i, y_i \le n$), meaning that on day i, a new directed road from x_i to y_i is built.

The next line contains two integers q and w ($1 \le q \le n^2$, $1 \le w \le 10^9$) — the number of wishes and the desired distance.

Each of the next q lines contains two integers s_i and t_i ($1 \le s_i, t_i \le n$) — the start and end of the i-th wish.

Output

For each wish, output a single integer — the earliest day that the wish can be fulfilled (i.e., after the road built on that day, there exists a walk of length exactly w from s to t). If it is never possible, output -1.

Example

standard input	standard output
6 3 1 3 2 3 3 1 4 4 5 5 5 2 5 5 6 5 8 5 1 5 5 1 1 4 5 4 6	-1 2 0 2 4

Note

In the sample, the desired distance is $w = 8$.

- For the first wish, it can be proved that it is impossible.
- For the second wish, after the roads $4 \to 5$ and $5 \to 5$ are built, one can walk along the town sequence $[5, 5, 5, 5, 5, 5, 5, 5, 5]$ to reach the destination.
- For the third wish, initially one can walk along the town sequence $[1, 3, 1, 3, 1, 3, 1, 3, 1]$ to the destination.
- For the fourth wish, after the roads $4 \to 5$, $5 \to 5$ are built, one can walk along the town sequence $[4, 5, 5, 5, 5, 5, 5, 5, 5]$ to the destination.
- For the fifth wish, after the roads $4 \to 5$, $5 \to 5$, $2 \to 5$, $5 \to 6$ are built, one can walk along the town sequence $[4, 5, 5, 5, 5, 5, 5, 5, 6]$ to the destination.